Skills Summary

Choosing and Combining Factoring Techniques

Use this reference while completing your worksheet or when studying.

Skill 1 — Take out the common factor first

Ask three questions, in this order: (1) Is there a factor in *every* term? Take it out. (2) How many terms are left — two or three? (3) Which pattern fits? Two terms with a minus between squares is a difference of squares; three terms is a trinomial.

Example: Factor $10x^2 - 15x$ completely.
Step 1. Question 1 first: both terms carry a factor, so the common factor comes out before anything else is considered.
Step 2. The greatest common factor of $10x^2$ and $15x$ is $5x$, and $10x^2 = 5x(2x)$, $15x = 5x(3)$.
$10x^2 - 15x = \boldsymbol{5x(2x-3)}$

Try it: Factor $8x^3 + 12x^2$ completely.

check: $\boldsymbol{4x^2(2x+3)}$

(!) Watch out: "completely" means keep going. $4x^2 - 36 = 4(x^2 - 9)$ is not the end — the leftover is a difference of squares and factors again.

Skill 2 — Trinomials that start with a plain x^2

Rule: $x^2 + bx + c = (x+p)(x+q)$ where $p \cdot q = c$ and $p + q = b$.

Signs: if c is positive, p and q share the sign of b; if c is negative, they have opposite signs.

Example: Factor $x^2 + 9x + 20$.
Step 1. Three terms, no common factor, and the leading coefficient is 1 — that is the product-and-sum method, not grouping.
Step 2. Product 20 and sum 9: the pair is 4 and 5, both positive because the product is positive and the sum is positive.
$x^2 + 9x + 20 = \boldsymbol{(x+4)(x+5)}$

Try it: Factor $x^2 - 2x - 15$.

check: $\boldsymbol{(x-5)(x+3)}$

Skill 3 — Trinomials with a number in front

Grouping method for $ax^2 + bx + c$: find two numbers with product $a \cdot c$ and sum b; split the middle term into those two pieces; group the four terms in pairs and take a common factor out of each pair.

Example: Factor $3x^2 + 11x + 6$.
Step 1. Three terms, no common factor, and the leading coefficient is not 1, so the product-and-sum shortcut does not apply — use grouping.
Step 2. Product $3 \cdot 6 = 18$, sum 11: the pair is 9 and 2. Split and group: $3x^2 + 9x + 2x + 6 = 3x(x+3) + 2(x+3)$.

$3x^2 + 11x + 6 = \boldsymbol{(3x+2)(x+3)}$

Try it: Factor $2x^2 - 7x + 6$.

check: $\boldsymbol{(2x-3)(x-2)}$

Skill 4 — Difference of two squares

Rule: $a^2 - b^2 = (a-b)(a+b)$. Two terms only, both perfect squares, joined by a minus sign. A *sum* of two squares does not factor.

Example: Factor $x^2 - 100$.
Step 1. Two terms rather than three, both perfect squares, with a minus between them — that rules out every trinomial method and picks this pattern.
Step 2. $a = \sqrt{x^2} = x$ and $b = \sqrt{100} = 10$, so the factors are $(a-b)$ and $(a+b)$.

$x^2 - 100 = \boldsymbol{(x-10)(x+10)}$

Try it: Factor $36x^2 - 1$.

check: $\boldsymbol{(6x-1)(6x+1)}$